

SCRIBE: Consistent Distributed Environment Modeling for Communication-Limited Robotic Exploration

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Abstract—Autonomous exploration of the ocean and other communication-limited environments requires agents to construct models from sparse measurements while using those models to determine where to sample next. Distributed teams can extend the scale and responsiveness of this process, but their ability to learn collectively depends on exchanging information across intermittent, bandwidth-constrained networks. Observation sharing grows with mission duration and sensor dimensionality, while conventional model-state fusion can become overconfident when model correlations are unknown. We introduce SCRIBE, a fixed-dimensional distributed environment modeling process for consistent model assimilation whenever communication is available. SCRIBE represents a scalar field through Gaussian basis functions and exploits the resulting linear information form to combine Information Kalman filtering, Covariance Intersection, and Metropolis–Hastings consensus. Agents can thereby exchange learned model states over any available peer link, without knowledge of the full communication graph or waiting for scheduled model-update windows. We evaluate SCRIBE with independent Vulcan information-based planners in twelve simulated four-robot trials under distance-limited communication and a complete blackout. The resulting trajectories achieve pooled posterior variance within 3.2% of ideal centralized sharing, indicating that consistent model assimilation can preserve effective distributed exploration through communication loss.

Index Terms—autonomous exploration, distributed estimation, informative path planning, intermittent communication, multi-robot systems

I. INTRODUCTION

Autonomous exploration is both a learning and a decision problem. A robot entering an unfamiliar region must gather observations, construct a model of the environment, and use that model to determine which future observations will be most valuable. Model confidence validates what has been learned and identifies where understanding remains incomplete. Gaussian Processes and related uncertainty-aware models have consequently found broad use in adaptive sampling [1] and information-based path planning [2].

The need for this capability is particularly clear in the deep ocean. An autonomous underwater vehicle may be asked to localize a hydrothermal plume, characterize a cold seep, or follow a chemical gradient using only the sparse measurements gathered during its mission; autonomous cold-seep exploration is one demonstrated example [3]. A team can observe a larger region and pursue several scientific targets concurrently, but underwater acoustic communication is intermittent, bandwidth-limited, and strongly dependent on vehicle separation [1]. Each robot must therefore continue learning and planning without contact, then incorporate its

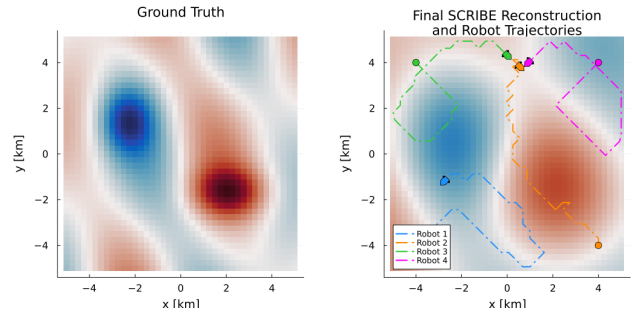


Fig. 1. Simulated scalar field over a 10 km $\times$ 10 km domain (left) and one robot’s final SCRIBE posterior mean (right). Large circles mark initial locations; dash-dot paths terminate in oriented robot symbols. The displayed model is the online posterior at the final sampling round, not a reconstruction formed by replaying the observation history.

peers’ learned model states whenever communication resumes.

Distributed estimates must remain probabilistically consistent. Existing work shares gathered or synthetic observations [4], [5], or distributes Gaussian-process model updates [6]. Observation sharing grows with the history and becomes especially burdensome for high-dimensional camera, sonar, or spectrometer records. Learned states are compact, but states that have interacted are correlated; treating them as independent repeatedly counts the same observations and overstates certainty [7]. That failure directly affects exploration because an overconfident model makes an unobserved region appear less informative than it is.

Finite-dimensional approximations reduce the cost of distributed Gaussian regression [8]. Related systems couple distributed models to informative planning [9], [10] or target search [11]. Together, these results motivate a compact model-learning backend whose confidence remains consistent as communication changes.

Signal-aware Collaboration for Reliable Information-based Estimation (SCRIBE) makes three contributions. First, an Information Kalman Filter (IKF) makes observations additive over Gaussian scalar-field (GSF) coefficients. Second, Covariance Intersection (CI) reconciles divergent priors without their unknown cross-correlations [7], [12]. Third, Metropolis–Hastings Markov Chain (MHMC) weights distribute new information using only the identities and degrees of presently available neighbors [13]. Following the hybrid consensus–CI construction for unreliable networks [14], SCRIBE permits model fusion whenever a peer link becomes available, without a full-graph view or a scheduled team-wide update.

II. DISTRIBUTED ENVIRONMENT MODELING

A. Fixed-Dimensional Information State

SCRIBE represents a scalar field through n_ϕ shared GSF bases [8], [15],

$$\mathcal{F}(x, k) = \psi^T(x)\phi(k), \quad \psi_i(x) = \tau_i^{-1} e^{-\|x - \mu_i\|_2^2 / \sigma_i^2}. \quad (1)$$

Only the coefficient vector ϕ is estimated online. For $z_j = H_j\phi + v_j$, agent j maintains information matrix $Y_j = P_j^{-1}$ and vector $y_j = Y_j\hat{\phi}_j$. A measurement contributes

$$\delta I_j = H_j^T R_j^{-1} H_j, \quad \delta y_j = H_j^T R_j^{-1} z_j. \quad (2)$$

The exchanged information vector and matrix are fixed by n_ϕ , independent of mission duration; replaying k observations grows as $O(kd_z)$. Raw sensor products can be processed locally before entering the shared field model.

Algorithm 1 Distributed Scalar-Field Modeling at Agent j

Require: GSF parameters $\{\mu_i, \sigma_i, \tau_i\}_{i=1}^{n_\phi}$
1: initialize (y_j, Y_j) and $\hat{\mathcal{F}}_j(x, 0) = \psi^T(x)\hat{\phi}_j(0)$
2: **while** executing the sampling mission **do**
3: $(z_j, X_j) \leftarrow \text{GatherSamples}()$
4: $(y_j, Y_j) \leftarrow \text{DistributedUpdate}(\psi, y_j, Y_j, z_j, X_j)$
5: $\hat{\phi}_j \leftarrow Y_j^{-1} y_j$; $\hat{\mathcal{F}}_j(x) \leftarrow \psi^T(x)\hat{\phi}_j$
6: **end while**

B. Fusion under an Unstable Network

Current innovations are new evidence; priors can contain an unknown amount of shared history. SCRIBE deliberately treats them differently. CI chooses simplex weights that minimize the covariance of the fused neighbor priors without assuming their cross-correlations. In parallel, innovations are averaged with $\gamma_{jm} = [1 + \max(|\mathcal{N}_j|, |\mathcal{N}_m|)]^{-1}$. These MHMC weights require only current neighbor degrees, rather than knowledge of the global or future graph. After local convergence over a reachable component of size n_c ,

$$y_j^+ = \tilde{y}_j + n_c \bar{\delta} \tilde{y}_j, \quad Y_j^+ = \tilde{Y}_j + n_c \bar{\delta} \tilde{Y}_j. \quad (3)$$

CI prevents prior agreement from masquerading as a new observation; MHMC propagates genuinely new information. This separation permits repeated, peer-level fusion whenever links become available, rather than only during scheduled team-wide update windows.

Algorithm 2 SCRIBE Distributed Update at Agent j

Require: Model ψ ; state (y_j, Y_j) ; sample (z_j, X_j)
1: $(y_j^-, Y_j^-) \leftarrow \text{LocalPriorUpdate}(y_j, Y_j)$
2: $H_j \leftarrow \psi(X_j)$; form $(\delta y_j, \delta I_j)$ using (2)
3: initialize prior and innovation consensus states
4: **while** local fusion has not converged **do**
5: exchange model-state components and degree with neighbors
6: fuse neighbor priors using CI
7: average neighbor innovations using MHMC
8: **end while**
9: **return** $(\tilde{y}_j, \tilde{Y}_j) + n_c(\bar{\delta} \tilde{y}_j, \bar{\delta} \tilde{Y}_j)$

Computational scaling. Let b be the GSF basis count or retained EOF rank, N the agents, E the active links, and d_j agent j 's degree. Each peer message is $\Theta(b^2)$. MHMC

costs $O(d_j b^2)$ per agent and $O(E b^2)$ fleet-wide per round; dense prediction costs $O(b^3)$, and J_{CI} CI evaluations cost $O(J_{\text{CI}}(d_j b^2 + b^3))$ per agent. Thus bounded-degree fleets scale linearly with N , whereas complete communication has $E = \Theta(N^2)$. On a fixed connected graph, disagreement falls from e_0 to ϵ in $\lceil \log(e_0/\epsilon)/(-\log \rho) \rceil$ MHMC rounds for mixing factor $\rho < 1$; under changing connectivity, its expectation depends on the link process.

III. EXPERIMENTAL EVIDENCE

We first evaluate SCRIBE independently of planning using four agents on fixed, matched sampling paths. A continuous-sharing check reproduces the centralized IKF to numerical precision. We then restrict every communication opportunity to disjoint, distance-limited peer pairs, remove all links for the middle third of the experiment, and restore only the same limited-range communication. Across 30 simulated trials with a truth represented by the GSF model, SCRIBE reaches final RMSE 0.1170 ± 0.0114 , compared with 0.1007 ± 0.0121 for centralized sharing, 0.1175 ± 0.0115 for CI alone, 0.2881 ± 0.0204 for the IKF without CI, and 0.8136 ± 0.0175 without communication. CI prevents priors formed during the blackout from being treated as independent when links return, while the IKF innovation update recovers information that CI alone can only conservatively intersect.

To demonstrate SCRIBE's effectiveness for exploration, four robots sample a simulated field for 36 rounds under the same limited communication and blackout schedule. Five comparisons isolate the sharing mechanism. IKF-only sums new innovations without reconciling correlated priors; CI-only intersects local posteriors without summing independent innovations; limited-observation sharing sends the two newest unseen records over each available link. No communication retains only local data, while centralized sharing supplies every observation as an ideal bound. Every distributed method obeys the same range limit: the largest component contains two robots, and the middle twelve rounds contain no links.

We use Vulcan [16] in a single-agent implementation for simplicity. Each robot plans from its own current model, without joint allocation or collaborative tree search. This deliberately isolates the information reward supplied by SCRIBE to otherwise decoupled planners. Figure 1 shows the online reconstruction and sampling paths. In Fig. 2, SCRIBE's final per-agent RMSE is 0.523 ± 0.060 , compared with 0.458 ± 0.052 centrally, 0.564 ± 0.084 for CI alone, 0.730 ± 0.081 for IKF alone, 0.653 ± 0.067 for limited-observation sharing, and 1.041 ± 0.053 without communication.

To evaluate exploration separately from fusion error, we pool the samples selected under each regime and construct one common evaluation model. Differences then arise from where the decoupled planners chose to sample, not from how their online models fused information. The pooled RMSE is 0.455 for SCRIBE, 0.458 centrally, and 0.594 without communication. Integrated posterior variance, the information reward used during planning, is 0.0488 for SCRIBE, within

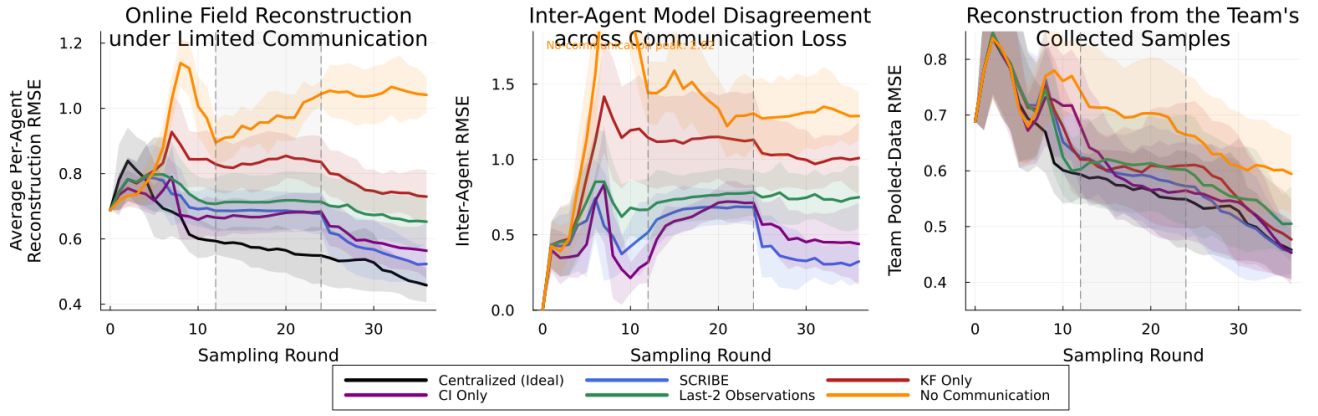


Fig. 2. Twelve simulated four-robot trials of 36 sampling rounds. Communication is distance- and bandwidth-limited to one in-range peer per robot in rounds 1–12 and 25–36; the shaded interval is a complete blackout. Curves show the mean and one standard deviation. The pooled-data panel isolates trajectory quality from fusion error. The disagreement axis is truncated to resolve the communicating methods; the annotation reports the no-communication peak. Lower values are better.

3.2% of the centralized 0.0473; CI alone reaches 0.0528, IKF alone 0.0617, limited-observation sharing 0.0651, and no communication 0.1074. SCRIBE therefore selects nearly as informative a set of measurements as ideal sharing, even though no distributed planner ever has access to a team-wide communication component.

SCRIBE communicates a learned state whose dimension is determined by b , rather than by mission duration or raw-sensor dimension. Although the simulated records are scalar, high-dimensional sonar, camera, or spectrometer products can be processed locally before entering the learned field model. Direct observation sharing must retain and transmit those records, a distinction that grows throughout the mission.

IV. CONCLUSION

SCRIBE provides a compact and probabilistically consistent environment model for distributed exploration under limited communication. Fixed-path results show that its hybrid update preserves consistency through disconnection and reconnection; planning results show nearly centralized information collection without a team-wide connected network. The communicated state remains $O(b^2)$, bounded-degree fleet work scales linearly with the number of agents, and local models remain usable throughout a blackout.

Because its update is independent of the planning architecture, SCRIBE can serve fully distributed GP and MCTS systems [9] and coordinated planners such as Multi-Agent Vulcan [17]. The same requirements arise in ocean, planetary, subterranean, and other remote missions where neither teammates nor human operators are continuously reachable.

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